Inherited variation

9A Genetics and evolution · Lesson 2 of 7

Date _____ Class _____

Today I can

- ☐ I can give examples of inherited variation and explain how sexual reproduction causes it.
- ☐ I can sort inherited variation into continuous and discontinuous, and recognise a normal distribution.

What happens today

1. Do Now: a litter of kittens
2. Learn: Inherited variation · Where inherited variation comes from · Inherited variation can be continuous or discontinuous · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 8 to 9 · Workbook pages 5 to 6

What I learned today (one or two sentences)

Key Vocabulary

Term	Definition
inherited variation	differences in characteristics that are passed on from parents to offspring
offspring	the young produced by parents
genetic information	the instructions for inherited characteristics, stored in the nucleus
gene	a section of DNA that carries the instructions for one characteristic
nucleus	the part of a cell that contains the genetic information
gamete	a sex cell (a sperm cell or an egg cell)
fertilisation	when a male gamete and a female gamete fuse (join together)
zygote	the fertilised egg cell
continuous variation	variation that can have any value between two points
discontinuous variation	variation with only a limited set of possible values
normal distribution	the bell shape you get when grouped continuous data are plotted

Quick Review

a litter of kittens

1 Give two ways these kittens vary from each other.

2 Which of these differences did they inherit from their parents?

3 From last lesson: is a scar inherited or environmental variation?

Core 1 of 2

Everyone does these. Answer on the card or under it.

1 Complete the sentences using words from the box. Use each word once.

Word box: characteristics fertilisation fuse genetic inherited offspring parents zygote

The nucleus of a cell contains the _____ information that controls _____ characteristics.

When two gametes _____ (join together) during _____, they produce a _____ (fertilised egg cell).

The fertilised egg cell contains genetic information from both _____, so it has instructions for inherited _____ from both parents. The _____ will have some characteristics of each parent.

2 Tick one box in each row to show what causes the variation: genes, the environment, or both.

Characteristic	Genes	Environment	Both
being able to speak more than one language			
length of hair			
human height			
natural eye colour			
blood group			
a scar			

Core 2 of 2

Everyone does these. Answer on the card or under it.

3 Class survey. Record the class totals in the table.

a) Complete the table using the class results.

Characteristic	Group	Number of students
tongue rolling	can roll	
	cannot roll	
ear lobes	attached	
	free (hanging)	

b) Is tongue rolling continuous or discontinuous variation? Give a reason.

Challenge 1 of 3

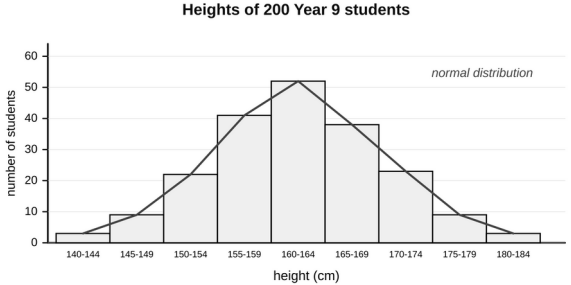
When Core is finished. Full sentences.

C1 Suggest a variation in tomatoes that is caused by both genetic information and environmental factors. Explain your reasoning.

Challenge 2 of 3

When Core is finished. Full sentences.

C2 The chart shows the heights of 200 Year 9 students.



a) Describe the shape of the chart and name this type of distribution.

Challenge 3 of 3

When Core is finished. Full sentences.

C2 *continued*

b) Explain why height gives this shape but blood group cannot.

c) Suggest one other characteristic that you would expect to show a normal distribution.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 Where in a cell is genetic information stored?

2 Name the cell formed when two gametes fuse.

3 Blood group: is it continuous or discontinuous, and inherited or environmental?

Today I can: Goal 1 ☐ Goal 2 ☐